Spectral Filtering for Vision–Language–Action Models

Shivam Kak [sk3686@princeton.edu] [1] Ishan Saha [is1893@princeton.edu] [1] Connor Brown [cb6856@princeton.edu] [1]

[1]Princeton University

PRINCETON UNIVERSITY

Abstract

We integrate Spectral Transfer Units (STUs) [2] into the $\pi_{0.5}$ flow-matching Vision–Language–Action model [1] and evaluate them on LIBERO [5] action prediction across *four* architectural placements: post-hoc on the action expert; pre-input on the noisy action chunk; on the predicted velocity; and pre-input on the PaliGemma prefix. We compare against a parameter-matched diagonal-SSM (Mamba) control to isolate Hankel-specific structure. Across all placements, all spectral-filter counts K, all action horizons $H \in \{5, 10, 20, 50\}$, and both warmup schedules, STU ties or underperforms the baseline. Mamba ties STU exactly, indicating the Hankel eigenbasis is not buying anything beyond generic linear recurrence in this regime. A short-warmup probe rules out the budget-bound hypothesis: forcing the spectral block to grow degrades the model. The result is a comprehensive negative finding that delineates where Hankel-eigenbasis filtering does and does not transfer.

Background: $\pi_{0.5}$ + Spectral Filtering

$\pi_{0.5}$ flow-matching VLA. A SigLIP vision encoder + PaliGemma language model + a small action expert. An action chunk $a \in \mathbb{R}^{H \times D_a}$ is produced by integrating a learned velocity field

$$x_{t+\mathrm{d}t} = x_t + \mathrm{d}t \cdot v_\theta(x_t, t), \quad x_0 \sim \mathcal{N}(0, I).$$

The action expert sees the full chunk in parallel, so causal attention is the only mechanism mixing information across the H time positions of the chunk—a natural fit for spectral methods.

Spectral Transfer Units (STU). Convolve along the horizon with eigenvectors ϕ_k of the Hankel matrix $Z_{ij} = \frac{2}{(i+j)^3 - (i+j)}$, scaled by the fourth root of their eigenvalues:

$$y = \sum_{k=1}^{K} \Big((u * \phi_k^+)\, M_k^+ \;+\; (u * \phi_k^-)\, M_k^- \Big)$$

where $\phi_k^\pm$ split filters into even/odd-symmetric components, $M_k^\pm$ are learnable, FFT in $O(L \log L)$.

Mamba-style control. Parameter-matched diagonal SSM $\dot{x} = Ax + Bu$, $y = Cx$ with HiPPO-LegS init. Isolates whether *Hankel-specific* structure helps vs. generic linear recurrence (per midterm reviewer ask).

Architectural Placements

v1 (post-hoc, action expert): residual STU *after* the action-expert transformer, randomly initialized.
v2 (pre-input, action expert): STU on the noisy action chunk x_t *before* the transformer; zero-init out-projection so step 0 is bit-identical to baseline.
v3 (output velocity): STU on predicted $v_t \in \mathbb{R}^{B \times H \times 32}$, the natural axis for a "smoothness" prior.
v4 (PaliGemma prefix): mirror of v2 on the VLM side; STU on the prefix embeddings (∼968 image+language tokens) before the VLM forward, zero-init out-projection.

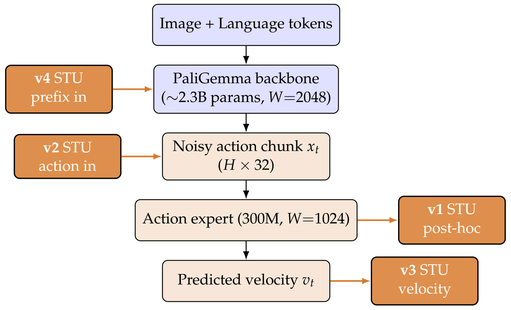

Figure 1. The four sites in $\pi_{0.5}$ where we tested an STU residual block. v1 is the original midterm design; v2/v3/v4 share the dreamer-aligned recipe (pre-input or pre-output, zero-init out-projection). Each placement isolates a different question: whether residual STUs help *after* a converged action expert (v1), at the chunk-input or velocity-output stages where horizon mixing is most plausible (v2/v3), or on the much larger vision-language backbone where capacity may be less of a bottleneck (v4).

Experimental Setup

Training. Fine-tune from $\pi_{0.5}$ *base* (`gs://openpi-assets/checkpoints/pi05_base`)—*not* the LIBERO-finetuned checkpoint. AdamW 5×10^{-5}, cosine decay (10k or 1k warmup), batch 8, gradient clip 1.0, bfloat16, 5000 steps per run.

Compute. **26 LIBERO fine-tunes**, ∼30 GPU-hours on Princeton's `della` cluster (H100 NVL 95 GB and 2× RTX PRO 6000 Blackwell 97 GB).

Evaluation. Last-100-step mean training loss (smoothed flow-matching MSE) is the reported metric; differences below ±1% are within run-to-run noise.

Results: H=10 Architecture Comparison

Model	Final Loss	Δ Base	Params
Baseline	**0.0299**	—	0
STU-K4 (post-hoc)	0.0316	+5.7%	8.4M
STU-K8 (post-hoc)	0.0313	+4.7%	16.8M
Mamba-K4 (post-hoc)	0.0311	+4.0%	8.4M
Mamba-K8 (post-hoc)	0.0312	+4.3%	16.8M
STU-v2-K4 (act-input)	**0.0298**	**−0.5%**	8.4M
STU-v2-K8 (act-input)	**0.0298**	**−0.4%**	16.8M
STU-v3-K4 (output vel)	**0.0297**	**−0.6%**	0.07M
STU-v4-K4 (PaliGemma)	**0.0296**	**−1.2%**	33.6M
STU-v4-K8 (PaliGemma)	**0.0296**	**−1.0%**	67.1M

Table 1. Last-100-step mean loss at H=10. Post-hoc residuals *hurt*; all three zero-init pre-input placements (v2, v3, v4) recover baseline within run-to-run noise.

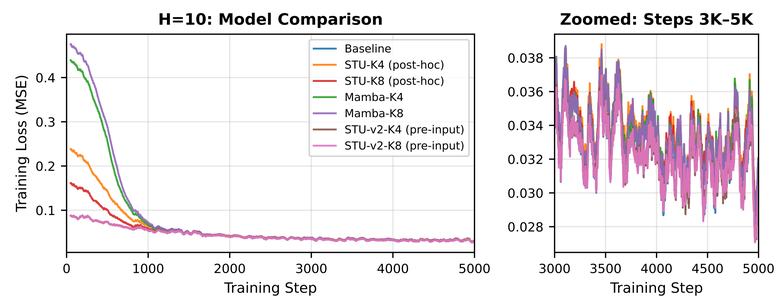

Figure 2. H=10 training loss (50-step rolling mean). Post-hoc variants (orange/green) start with elevated loss and only catch up by step ∼2000. Zero-init STU-v2 (brown) overlaps baseline (blue) from step 0—the visual signature of zero-init.

Horizon Ablation

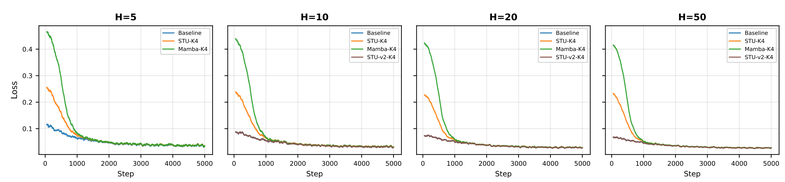

Figure 3. Horizon sweep ($H \in \{5, 10, 20, 50\}$). Post-hoc variants elevated at every horizon. STU-v2 (brown) overlaps baseline from step 0 at every horizon. Longer H reaches lower terminal MSE because each loss value averages more correlated time-steps.

H	Base	STU-K4 post-hoc	STU-v2-K4	STU-v4-K4
5	0.0338	0.0352 (+4.2%)	—	—
10	0.0299	0.0316 (+5.7%)	**0.0298**	**0.0296**
20	0.0276	0.0286 (+3.5%)	**0.0275**	**0.0275**
50	0.0269	0.0274 (+1.9%)	**0.0270**	—

Table 2. Horizon ablation, all warmup-10k. Post-hoc gap shrinks (+5.7% @ 10 → +1.9% @ 50) but does not close. Linear extrapolation crosses zero only at $H \approx 80$–100, beyond LIBERO. Zero-init v2 & v4 tie baseline at every horizon tested.

Spectral Weight Analysis

We loaded the v2 checkpoints and inspected $M_k^\pm$ directly to verify the spectral block is being trained:

$$\max |M_k^\pm| \approx 8\times10^{-3}, \quad \overline{|M_k^\pm|} \approx 7\times10^{-4}$$

Frobenius norms 0.29–0.49 across the full $K \times D_a \times W$ tensor. Default random-init scale would be $(KD_a)^{-1/2} \approx 0.088$, so after 5k steps $M_k^\pm$ has grown to about **1% of a typical layer's magnitude**. The optimizer has measurably moved $M_k^\pm$ off zero, but not enough to materially alter the network's output.

Warmup Schedule Sensitivity

Shortening LR warmup from 10k to 1k steps causes $\|M^+\|_F$ to grow from 0.36 to 0.92 (2.5×), forcing the spectral block off zero-init faster.

	Base	v2-K4	v3-K4	v4-K4
warmup-10k	0.0299	−0.5%	−0.6%	−1.2%
warmup-1k	0.0277	+1.5%	+2.0%	+1.9%

Table 3. Warmup probe at H=10, all three zero-init placements. Absolute loss improves with shorter warmup at every site, but the relative position of every STU variant *worsens* when $M_k^\pm$ is allowed to grow—consistent across all three placements. **This rules out the budget-bound explanation:** more learning signal → *worse*, not better.

Key Findings

1. **Negative result is robust across 4 placements** (post-hoc, action-input, output-velocity, VLM-prefix), all K, all H, both warmups. Spectral filtering does not add a useful inductive bias for $\pi_{0.5}$ action prediction on LIBERO.
2. **Hankel structure is irrelevant here.** Mamba ties STU at every configuration tested, so the Hankel eigenbasis is not buying anything beyond generic linear recurrence.
3. **The horizon hypothesis is directionally right but quantitatively weak.** The post-hoc gap shrinks with H, but linear extrapolation crosses zero only at $H \approx 80$–100, beyond LIBERO.
4. **Placement and zero-init matter more than the spectral basis.** The pre-input zero-init recipe rescues the negative result everywhere (no longer hurts), but never produces a positive one.
5. **Even on the much larger PaliGemma backbone (v4),** with 25× more parameters than the action expert, the result is unchanged. The headroom hypothesis is not the explanation.

Discussion

Why doesn't the spectral block engage? The dreamer-aligned recipe makes the residual branch start at zero, but it does not give it any reason to leave zero. With H=10 noisy chunks predicted from a near-converged backbone, the per-step MSE gradient already lives in the action expert's existing weight space; the optimizer has no incentive to route it through a freshly-initialised parallel pathway. The 1k-warmup probe makes this concrete: when we force $M_k^\pm$ to grow 2.5×, performance *worsens* at every placement.

The horizon hypothesis is real but small. The post-hoc gap shrinks monotonically from +5.7% at H=10 to +1.9% at H=50, validating the directional claim from our midterm. But linear extrapolation crosses zero only at $H \approx 80$–100, well beyond what LIBERO action chunks support. The spectral inductive bias has the right shape; the wrong magnitude.

Architecture matters, structure does not. Mamba-K{4,8} ties STU at every configuration, so the cost—and the benefit—of inserting a new linear-recurrence branch is dominated by *where* you put it and *how* you initialise it, not by the choice of Hankel eigenbasis vs. HiPPO-LegS diagonal SSM.

Future Work

- Success rates on the 130 LIBERO tasks (we measure only training MSE; rollout behavior could differ even at near-equal MSE).
- A generic-SSM control on the VLM side (Mamba-v4): the FFT-based diagonal-SSM kernel exceeds 32 GB on the 968-position prefix at W=2048; a chunked-FFT or selective-scan re-implementation would close this gap.
- STU-v3 at $H \in \{20, 50\}$ and STU-v4 at H=50 (compute-bound).
- A learned residual gate, to verify the optimizer is not actively zeroing the block.
- Pre-training a $\pi_{0.5}$ checkpoint *from scratch* with the spectral block in place, to test whether STU helps when not introduced as a residual on a converged backbone.

References

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